Space Operations Take Home Test

Space Systems Engineer

Anomaly detection

Our satellites produce a lot of data daily, both telemetry and logs. Parsing the right data in the right way allows us to recognize anomalies and automate responses.

We are sending you data from seven satellites in .csv format. The goal is to analyze this data to identify potential anomalies. The data contains telemetry readings of 9 of the sensors onboard the satellites, as well as the beta angle of their orbits.

You will be asked to present information clearly and to analyze it after. The result that you are expected to provide is a **.zip** file consisting of:

- A Python script
- A number of plots in a .png or .jpg format
- A text file (pdf or a simple .txt) providing answers to the questions that are asked

Some notes:

- Under nominal conditions, the satellites have 3 gyros and 4 wheels active, which get saturated at 8000rpm.
- The telemetry channel under the name “low_voltage” is a flag triggered by the satellite when it detects that the voltage of its bus is below a certain threshold. As a result, the satellite will inhibit some of its hardware such as the camera.
- Gyro rates are expressed in degrees per second, wheel speeds are in rpm, voltage is in volts
- Values that are empty or not a number are due to the lack of a measurement at that epoch
- The satellites are named following the phonetic alphabet (alfa, bravo, charlie...).

Special remarks:

- Try to keep your answers concise, a couple of sentences can be enough.
- As part of your code, include a print line outputting the **time** required to run in total.

Questions

1. Write a Python script to retrieve all the data from the .csv files, filter the data to make it presentable and plot the following (attach the plots to your deliverable):
    a. A plot per satellite including all telemetry channels
    b. A plot per telemetry channel including all satellites.
    c. The beta angle of all the satellites, combined in a single plot
2. For each satellite, compute the percentage of the available telemetry points where:
    a. One or more wheels are saturated
    b. The low voltage flag is active
    c. One or more gyro rates are above two degrees per second
3. In light of the results, provide an answer to the following questions
    a. The PlanetScope constellation is divided into “flocks” of satellites: batches of satellites launched at the same time by the same vehicle. How many flocks can you distinguish in these 7 satellites? How can you tell? Which satellites belong to each flock?
    b. Some satellites clearly show anomalies that could be dangerous for their operations, can you identify at least two of them?
4. For the anomalies identified, create a separate Python function that:
    a. Takes as input a telemetry channel.
    b. Returns a boolean, specifying if the issue in question is present (True) or not (False). There should be no false positives.